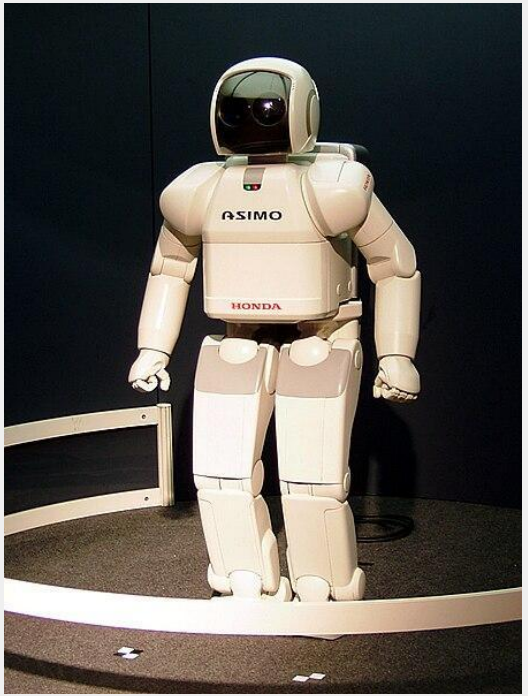
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 2 January 31, 2025

AI History

Since when, AI? + What is AI anyway?

Early Modern



Modern



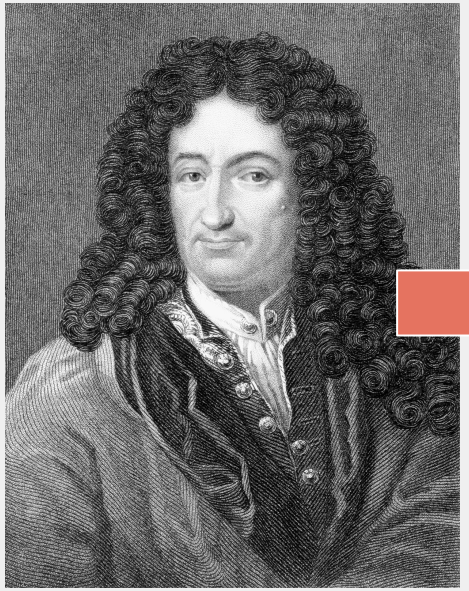
Large-Scale AI?



AI History

Since when, AI? + What is AI anyway?

Early Modern



Modern



Large-Scale AI?



This Week: Early Modern to Modern

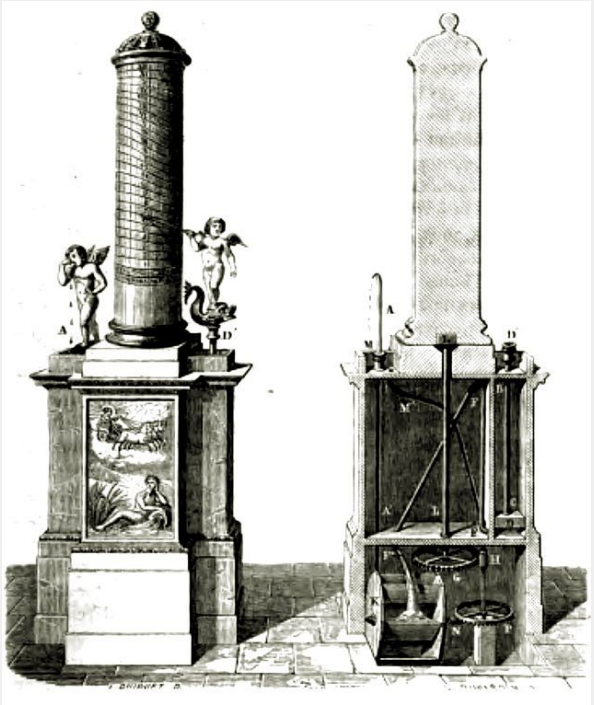
This Week: Early Modern to Modern

Next Week: “Modernization” & Cold War

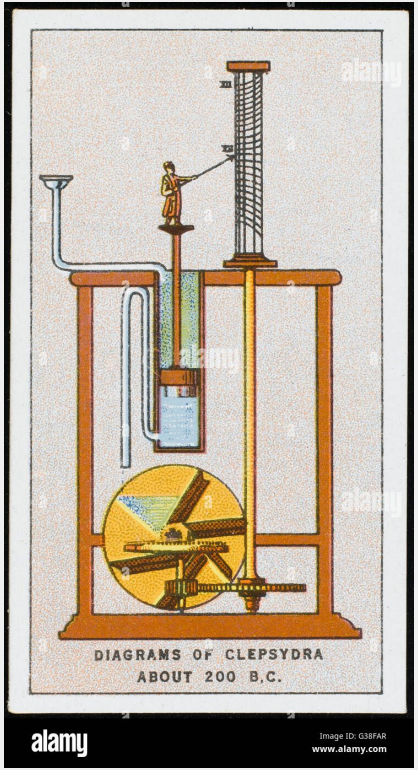
- What is automation?
- Why automation?
- Living things vs. Mechanical things
- Science & Faith

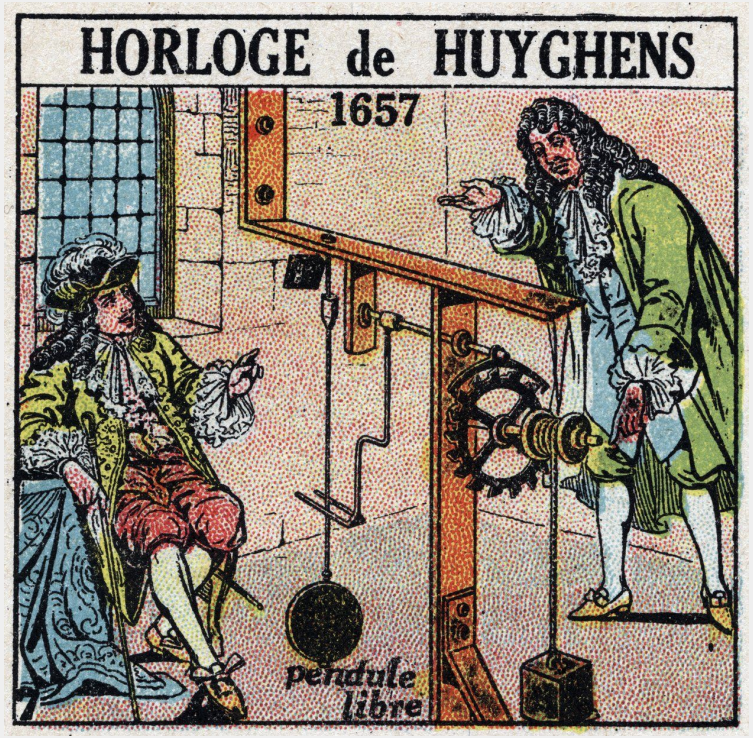
Ancient AI

How does this differ from the pendulum clock?



Ktseibios of Alexandria's Clepsydra. 250 BCE





1657 Pendulum Clock, Christian Huygens

Early Modern

16thC Protestant Reformation & Challenge the Pope
Mind-Body Separation



- Material entities have no agency (passive)
- Reformers asserted distinction between matter and spirit, setting their clockmaker God apart from his clockwork
- Only god has agency vs. medieval Catholic God had had presided over a cosmos permeated by spirit.

Agency

Organicist/Naturalist:

Nature = Agency = Life



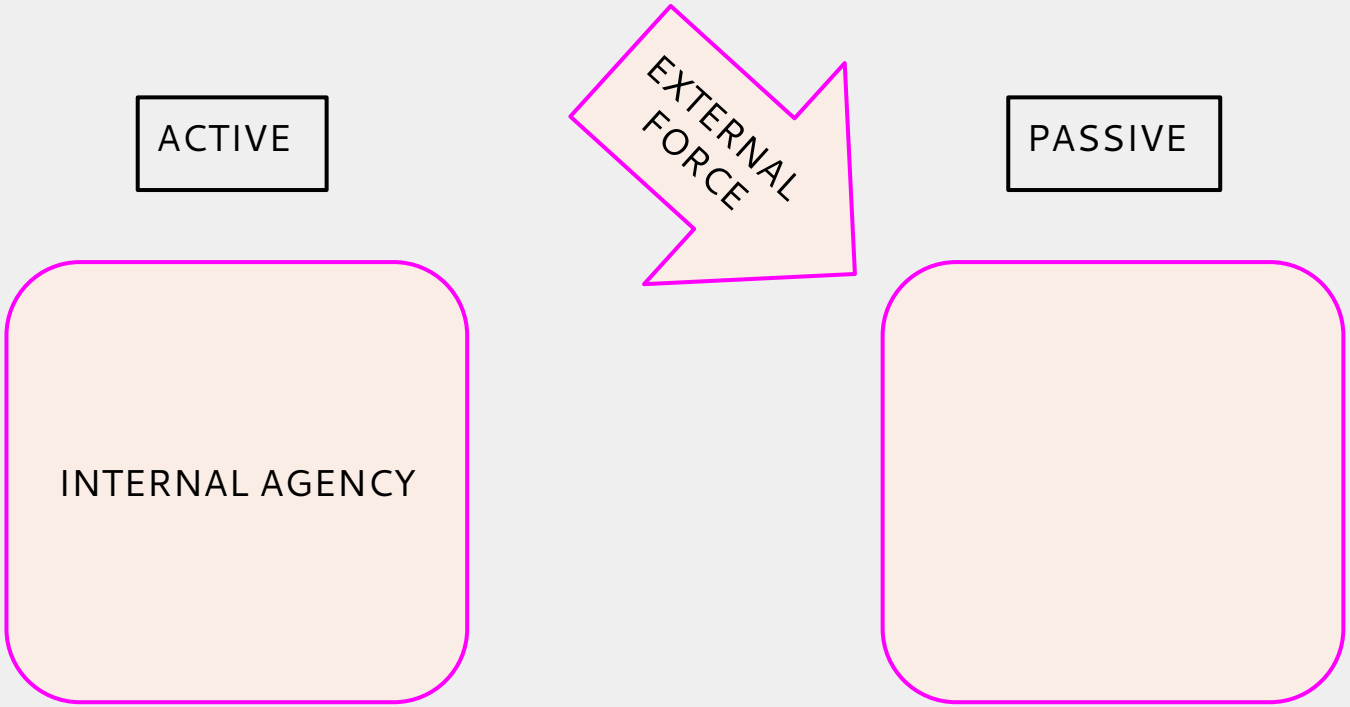
Mechanist:

Banish all agency from Nature

Everything can explained with physics

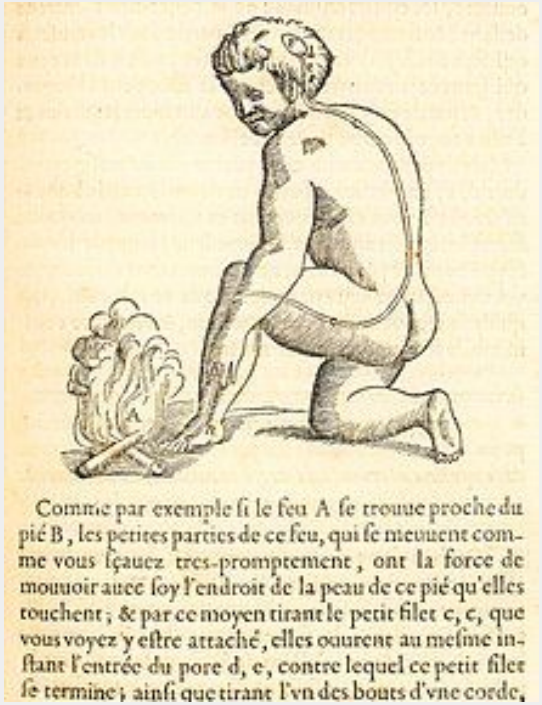
Early Modern

17thC Debates on “Agency” in Nature



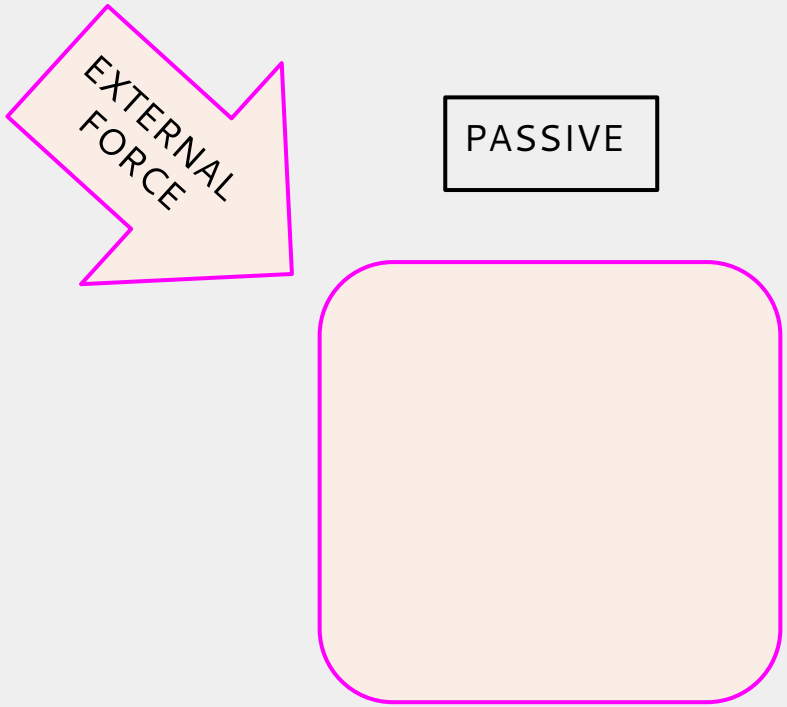
Passive Mechanist

Entirely devoid of agency



Comme par exemple ſi le feu A ſe trouue proche du pié B, les petites parties de ce feu, qui ſe meuuent comme vous ſçauez tres-promptement, ont la force de mouuoir auec ſoy l'endroit de la peau de ce pié qu'elles touchent; & par ce moyen tirant le petit filet c, c, que vous voyez y eſtre attaché, elles ouurent au meſme inſtant l'entrée du pore d, e, contre lequel ce petit filet ſe termine; ainſi que tirant l'vn des bouts d'vne corde,

Descartes's Traite de l'homme, 1664



Descartes
Newton
Boyle

Descartes

Passive, brute mechanism; “Classical mechanist”

Cartesian dualism—the idea that matter and spirit or soul were two absolutely distinct things

Only reason/rationality is from the soul (animals have no rationality)

One could explain every aspect of life in terms of machinery, and so could understand the workings of living beings as fully as a clockmaker understands a clock. Rather than to reduce life to mechanism, he meant to elevate mechanism to life: to explain life, never to explain it away.



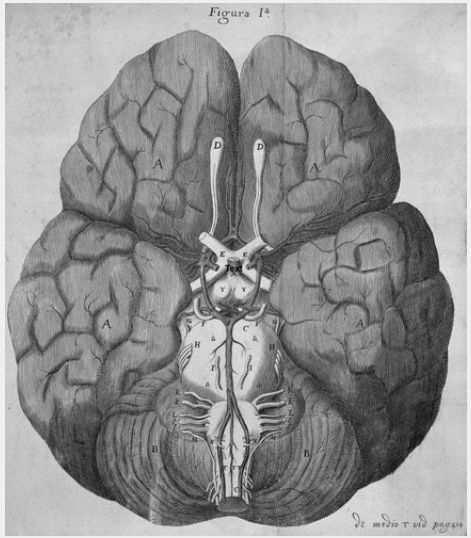
Active Mechanist

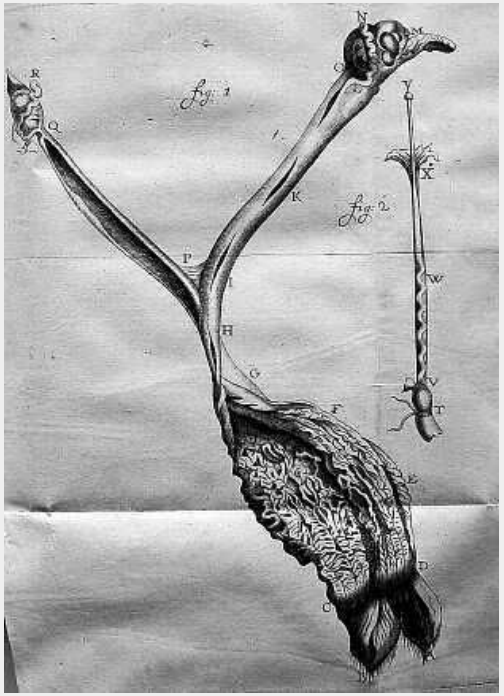
Actively Self-constituting

ACTIVE

INTERNAL AGENCY

Leibniz
Lamarck





Leibniz

Defines “Organism” through Active Mechanism

God works thru organization not design

- “God preformed things, such that the new organizations are but a mechanical consequence of a preceding organic constitution; as when butterflies come from silkworms.”

An animal was “a living organized body” whose identity attached to its organization. The identity of a human being consisted in an ongoing organization of ever-renewing material parts.

“I define the Organism, or the natural Machine.. a machine in which each part is a machine, and consequently the subtlety of its artifice goes to infinity, nothing being small enough to be neglected, whereas the parts of our artificial machines are not machines. That is the difference between Nature and Art.”

Mechanical “restless”? watches



Astronomic Watch 1660



Thomas Tompion’s Balance Spring Watch



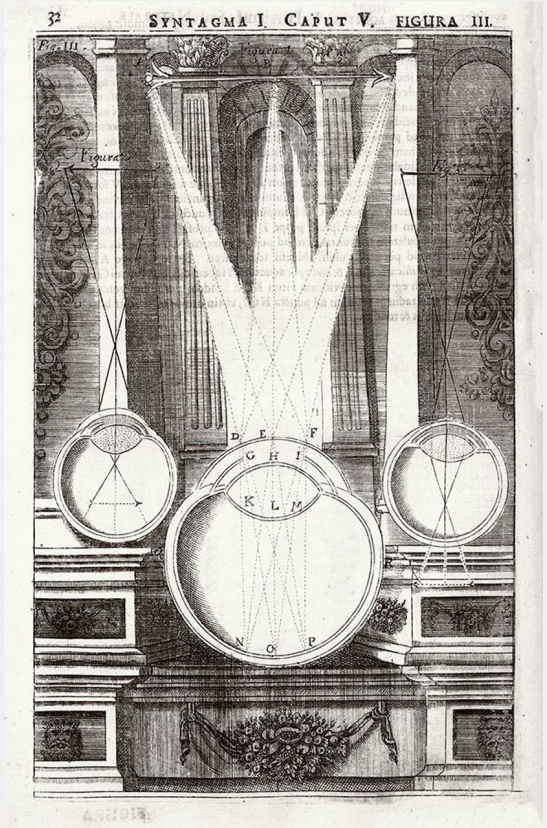
Portrait of Gentleman with Watch 1675

New Science's theological origins

"Nature is God's Artifact"

"There is no more rashness to say, that an Eye, than that a Telescope, was made for an Instrument to See with." - Boyle

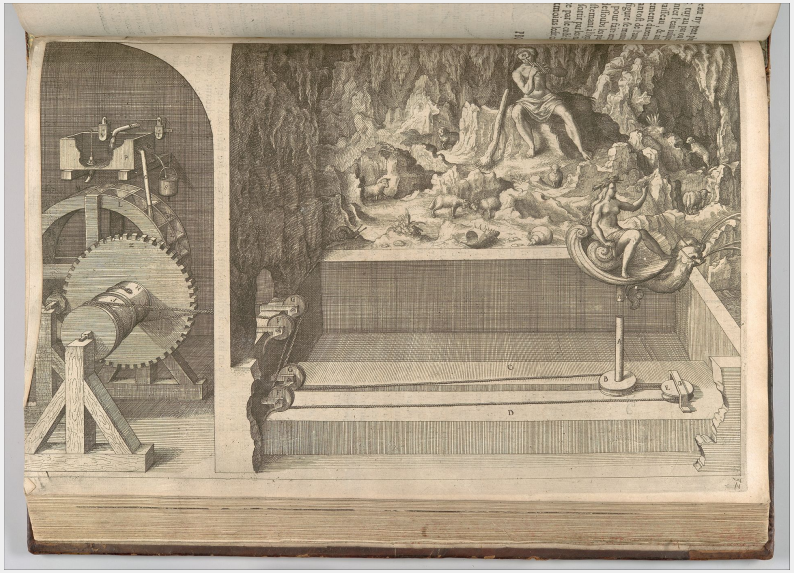
- Demonstrate God's existence by showing the detailed perfection of his world-machine



Oculus Artificialis, Johann Zahn. 1685

Animal-/Man- Machines

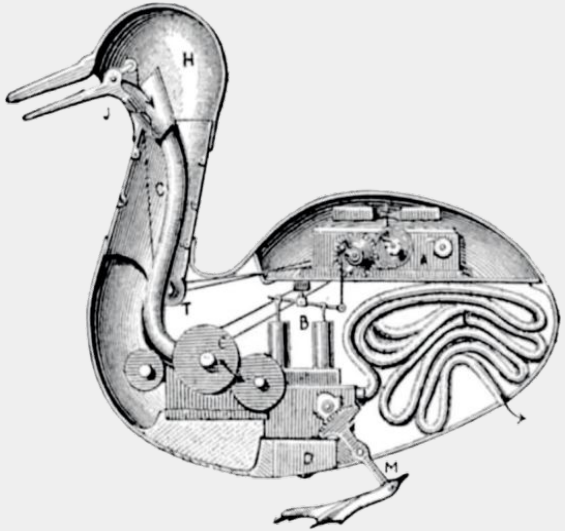
Hydraulic & Mechanical Engineering

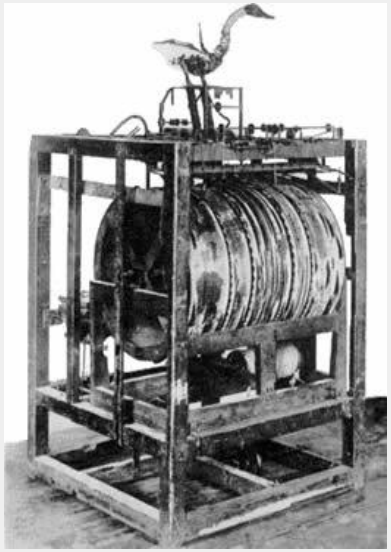


Les raisons des forces mouvantes,
Isaac De Cause. 1615

Enlightenment's first "androids"

"Prometheus's Rival"



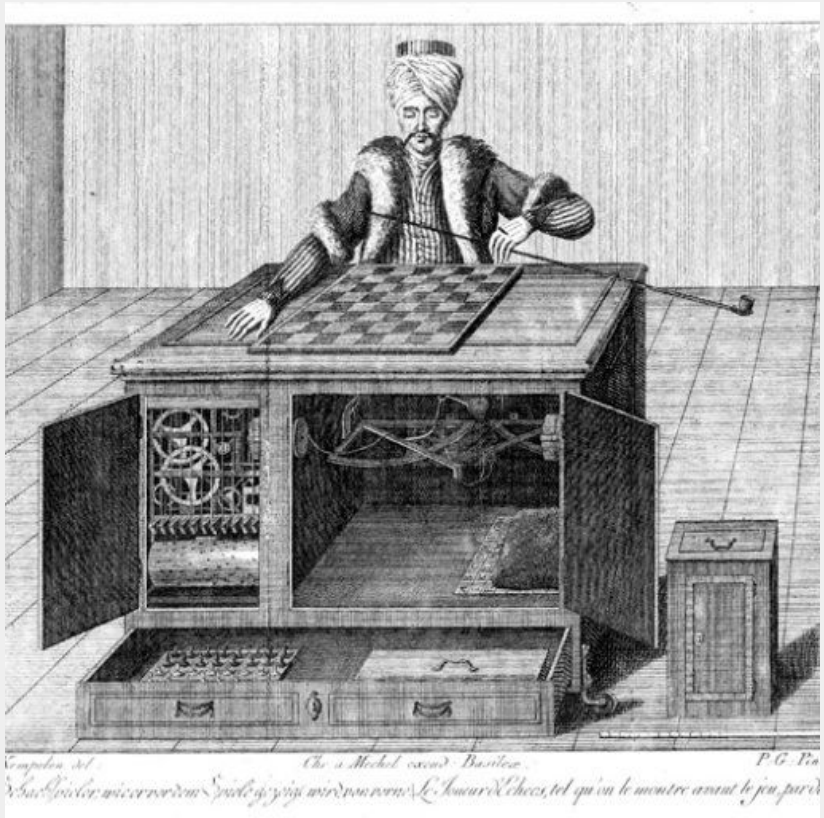


Canard Digérateur, Jacques Vaucanson. 1764



Mecanisme du fluteur automate, Vaucanson. 1738

Man-Machines

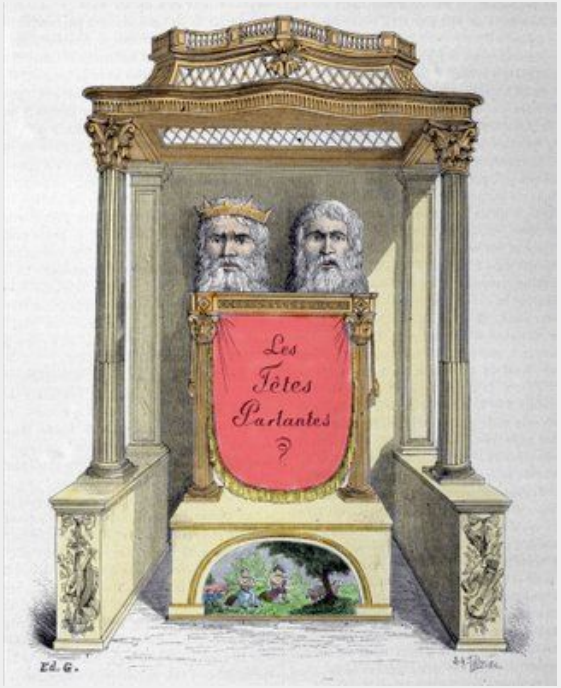


Chess-Playing Turk Wolfgang von Kempelen 1782

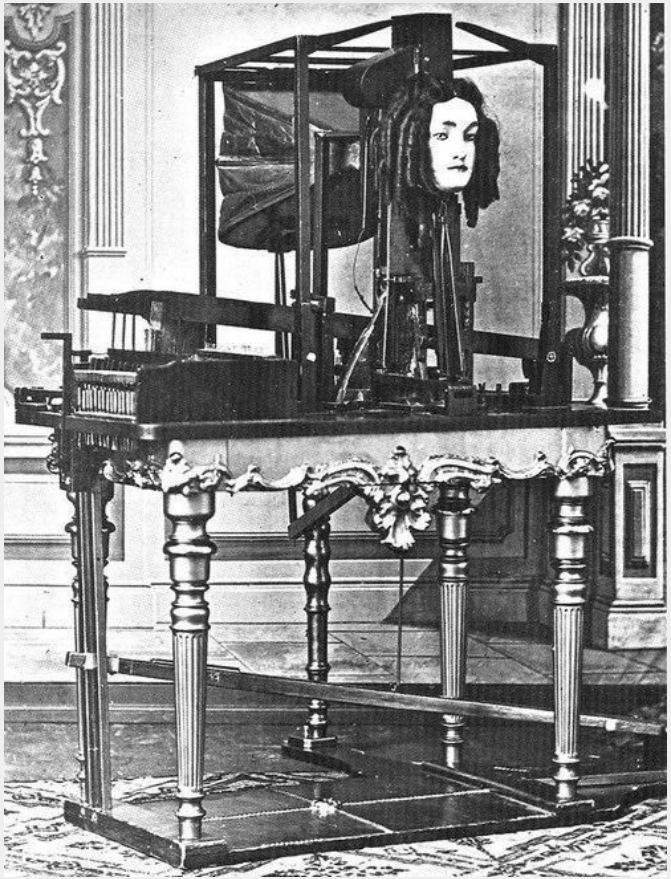
Speech Imitation

17th C obsession with “speech” autonoma

In praise of Louis XVI: “The King gives peace to Europe,” intoned the first head; “Peace crowns the King with Glory,” replied the second; “and Peace makes the Happiness of the People,” added the first; “O King Adorable Father of your People their Happiness shows Europe the Glory of your Throne”



abbe Mical discoursing heads 1778



Faber’s Talking Head 1846



Jacqet-Droz 3 automata 1772

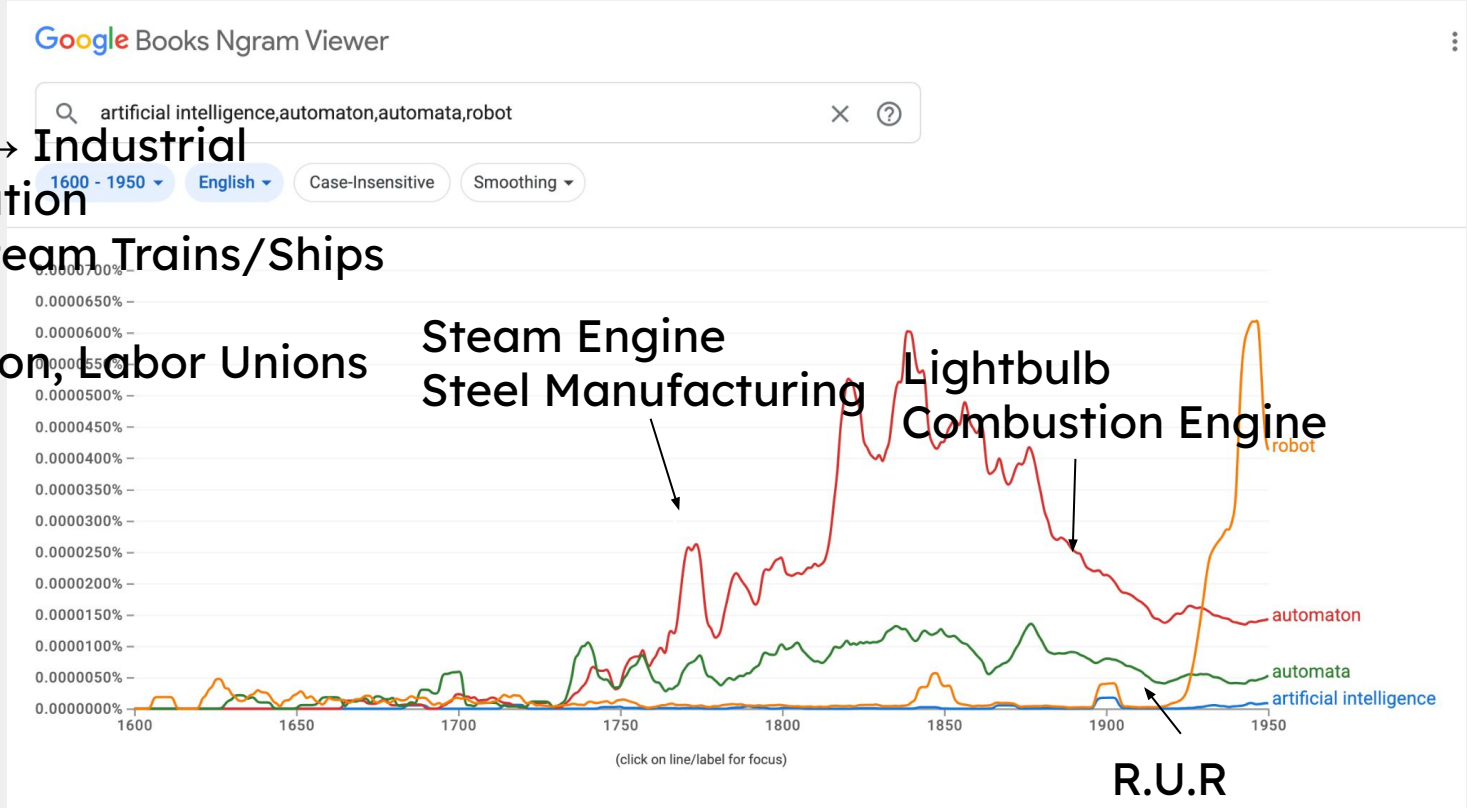


Karakuri Ningyo 17th C

- Similarities and differences
  - Are they both influenced by faith? (e.g. Japanese Shintoism)
  - Mimic “human” displays of self-will

1st Industrial Revolution

Agrarian → Industrial
Mechanization
Textiles, Steam Trains/Ships
Iron/Coal
Urbanization, Labor Unions



Punch Cards



Jacquard Loom 1804

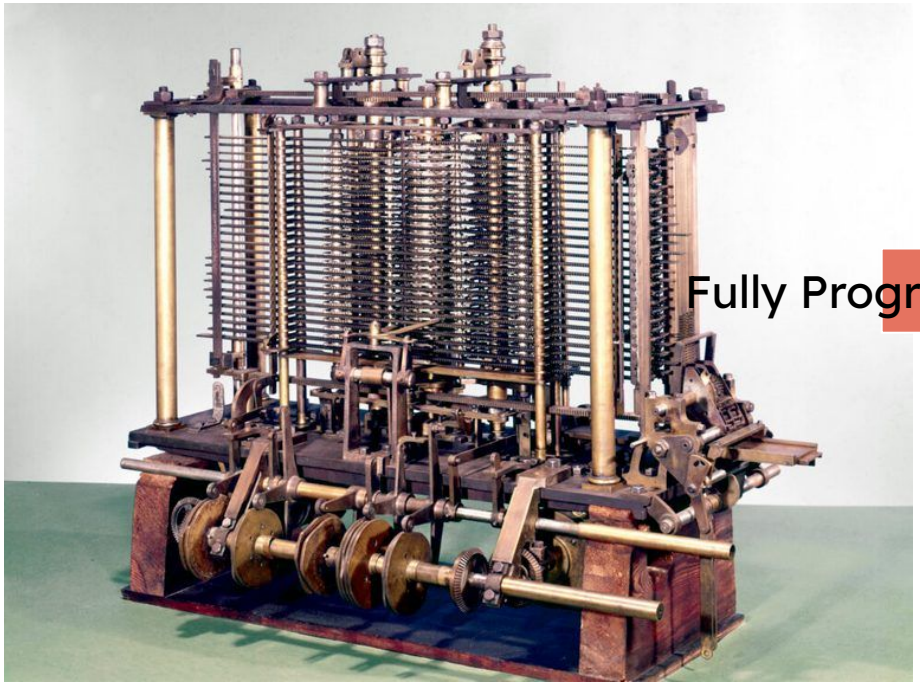


Jacquard woven on Silk. 1839



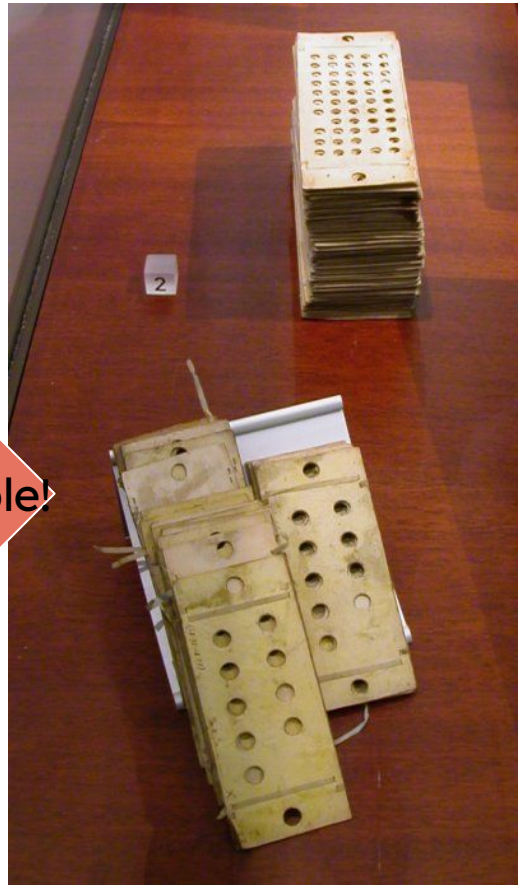


1820s Start of Computation



Difference Machine, Charles Babbage. 1820s

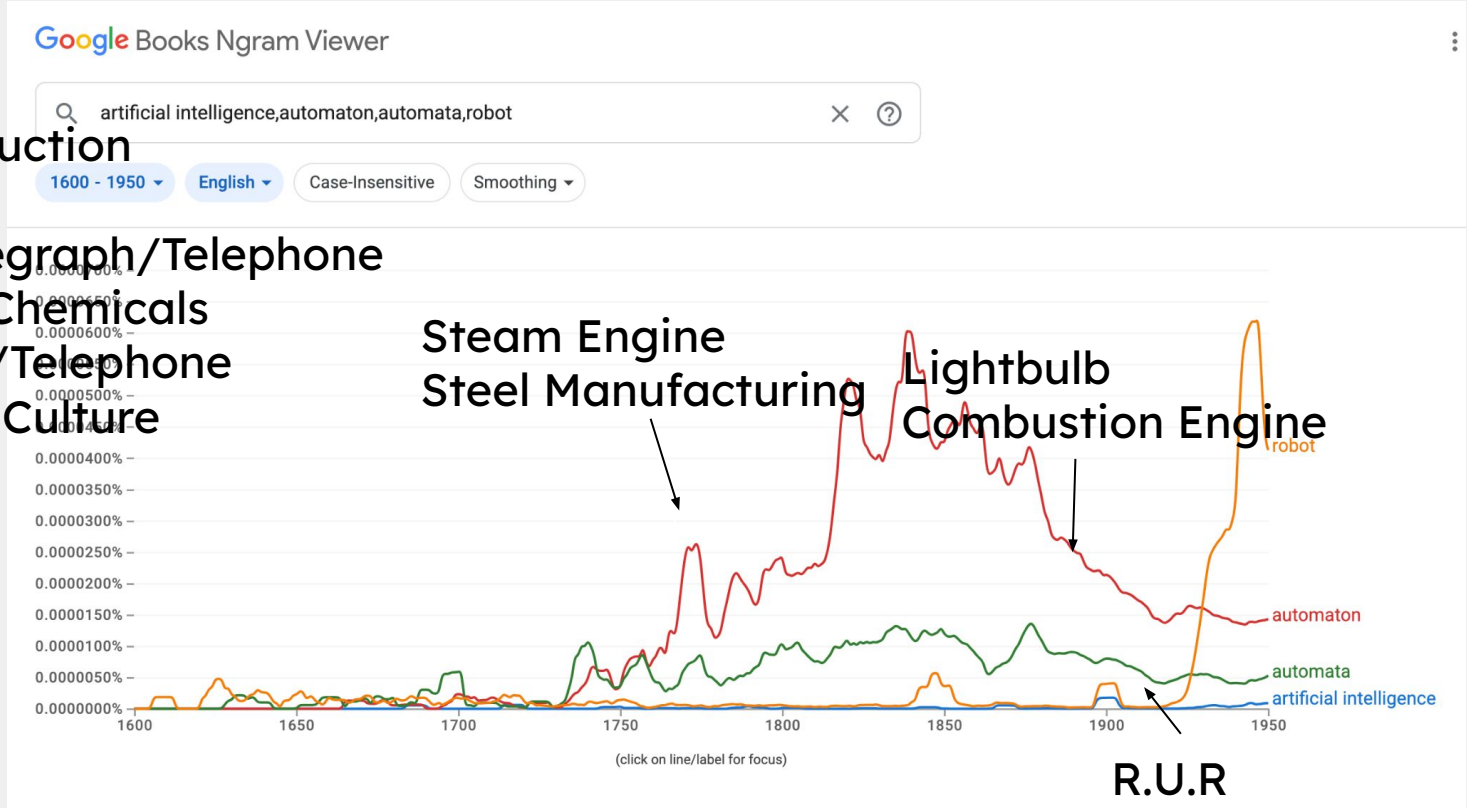
Fully Programmable!



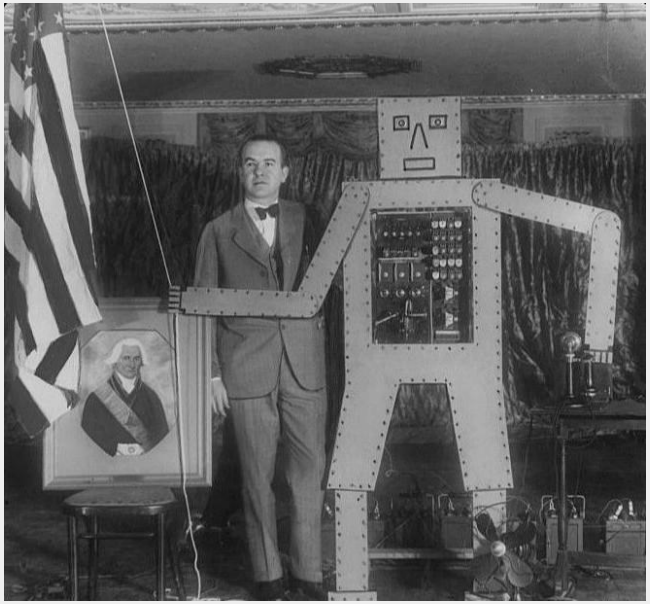
Punch Cards, Analytical Machine. Charles Babbage

2nd Industrial Revolution

Mass Production
Electricity
Radio/Telegraph/Telephone
Steel/Oil/Chemicals
Telegraph/Telephone
Consumer Culture



Man-Machines of 20th C



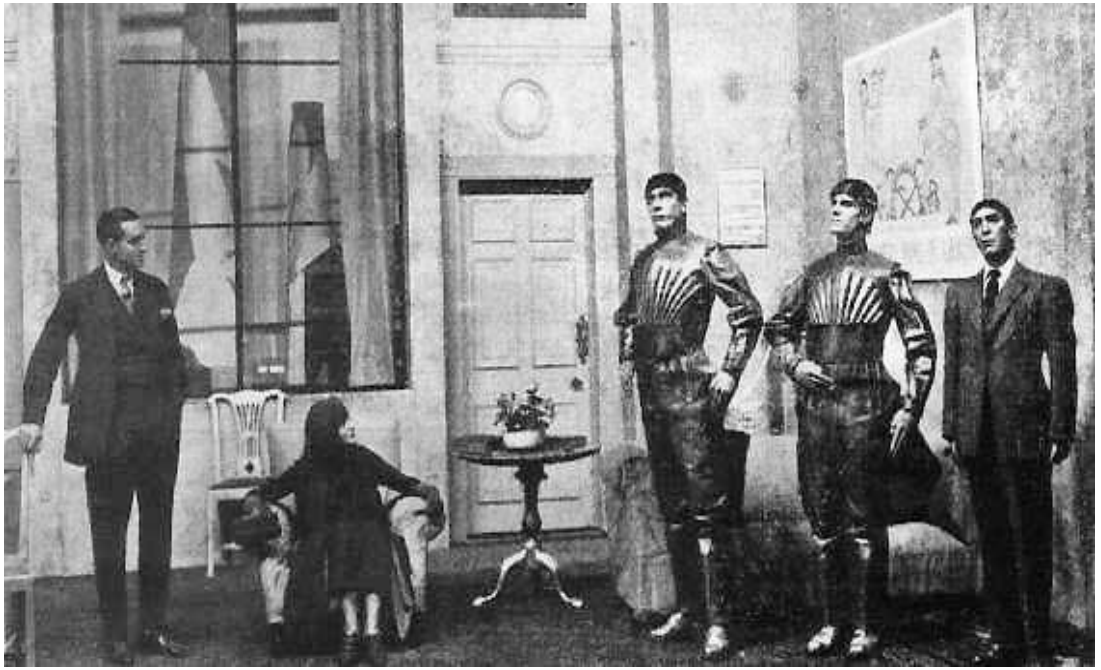
Mr. Televox. R.J. Wensley, Westinghouse (Westinghouse Electrical Company) 1926



Eric the robot at Model Engineer Exhibition at the Royal Agricultural Hall in London. 1928

“Robot” replaces “Automation” & “Android”





“Rossum’s Universal Robots,” Karel čapek. 1921

- Similarities and differences
  - Automation alacrity vs. fear
  - Tools vs. servants

Mid-20th C First digital computers
Human Mind/Intelligence ⟷ Computers

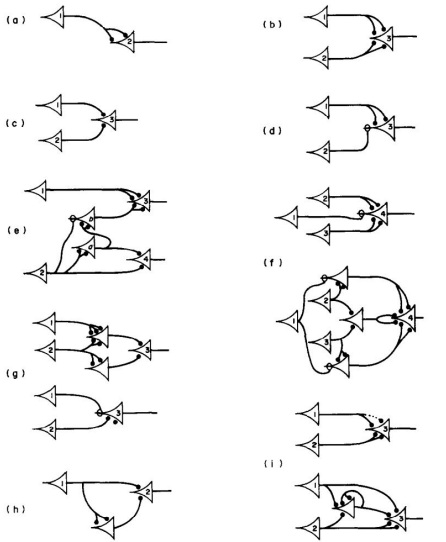
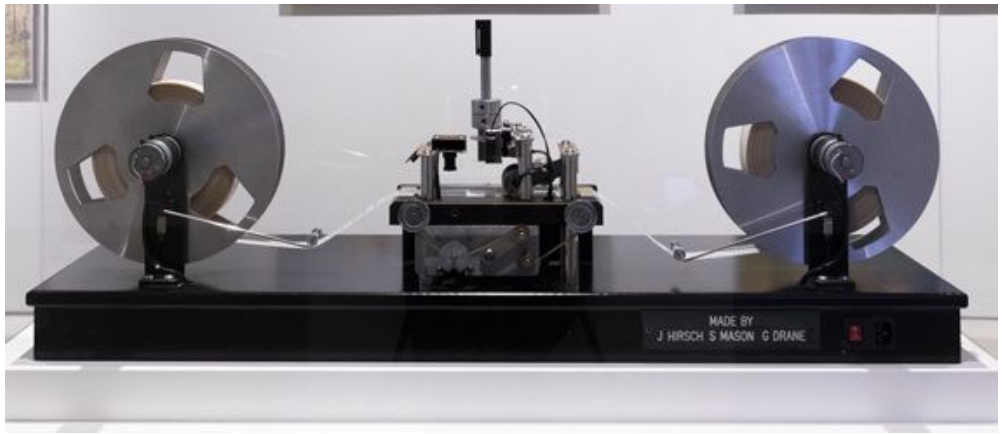


Figure 1. The neuron c_i is always marked with the numeral i upon the body of the cell, and the corresponding action is denoted by "N" with i s subscript, as in the text:

Pitts-McCulloch "Artificial Neural Network Theory." 1943



Physical model of Alan Turing's 1936 Turing Machine

Darwin

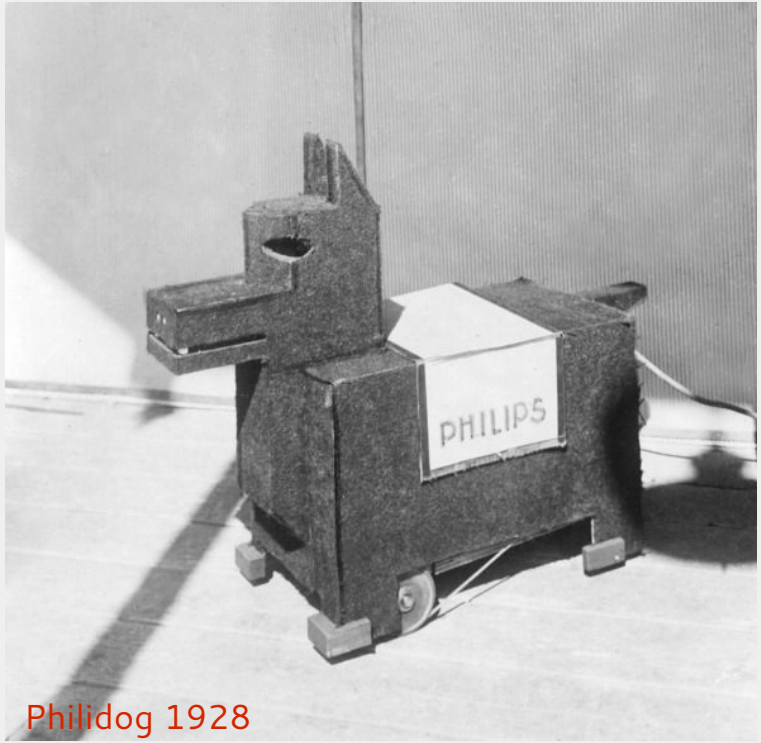
Straddling both active and passive views

Neither internal (striving)/external (divine) agency but:

1) living things actively improved themselves over time (historical agency)
2) living beings are passive, designed artifacts (divine agency)

> It is interesting to contemplate a tangled bank, clothed with many plants of many kinds, with birds singing on the bushes, with various insects flitting about, and with worms crawling through the damp earth, and to reflect that these elaborately constructed forms, so different from each other, and dependent upon each other in so complex a manner, have all been produced by laws acting around us. These laws, taken in the largest sense, being Growth with Reproduction; Inheritance which is almost implied by reproduction; **Variability from the indirect and direct [external removed] action** of the conditions of life, and from use and disuse: a Ratio of Increase so high as to lead to a Struggle for Life, and as a consequence to Natural Selection, entailing Divergence of Character and the Extinction of less-improved forms. Thus, from the war of nature, from famine and death, the most exalted object which we are capable of conceiving, namely, the production of the higher animals, directly follows. There is grandeur in this view of life, with its several powers, having been originally breathed by the ***Creator*** into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from so simple a beginning endless forms most beautiful and most wonderful have been, and are being evolved.
>
> From Origins of Species 1859

“Sensitive Automata”

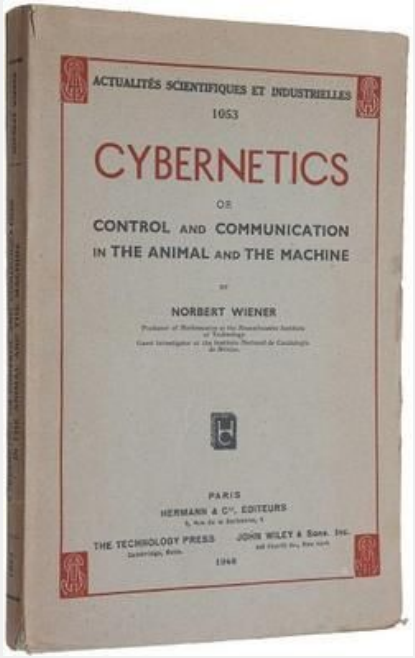


Philidog 1928



Cybernetics

“Agencies acting to maintain constancy”
Passive mechanist, neo-Darwinism

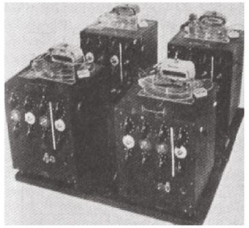


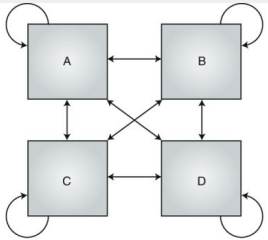
Norbert Weiner. 1949

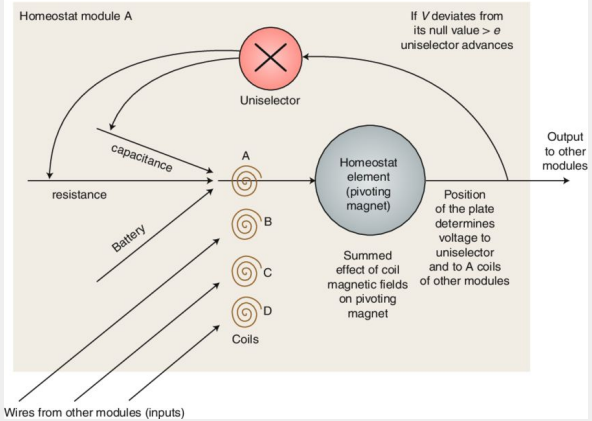
responsiveness is what makes organisms “living things” and responsiveness can be mechanically imitated → living organisms can be mechanically imitated

Cybernetics

Homeostasis and Negative Feedback







Homeostat with Negative Feedback.
Walter Ross Ashby. 1949



FIGURE 8/2/1: The Homeostat. Each unit carries on top a magnet and coil such as that shown in Figure 8/2/2. Of the controls on the front panel, those of the upper row control the potentiometers, those of the middle row the commutators, and those of the lower row the switches *S* of Figure 8/2/3.

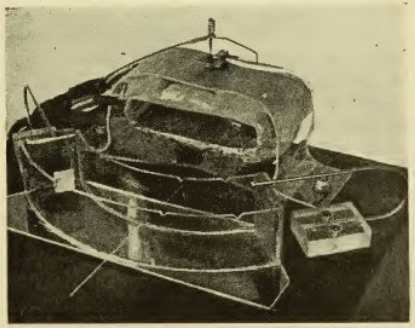
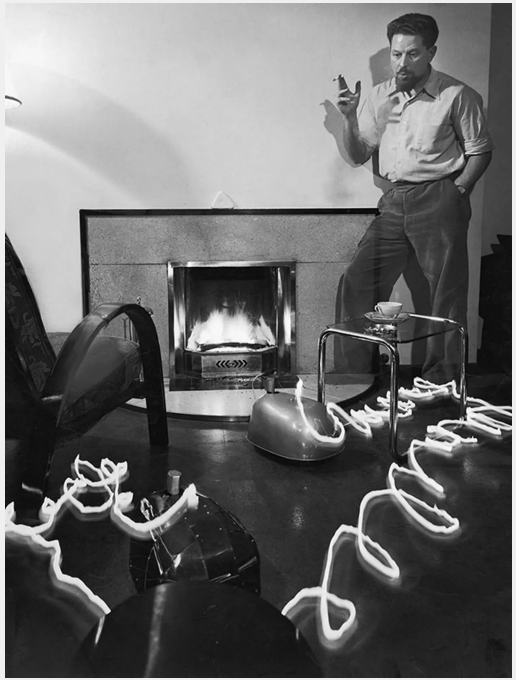
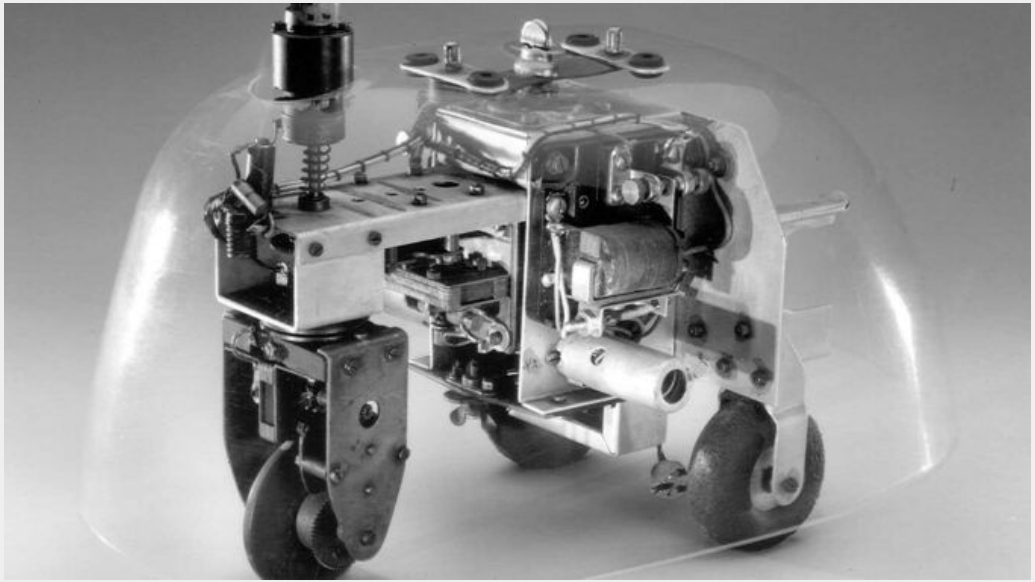


FIGURE 8/2/2: Typical magnet (just visible), coil, pivot, vane, and water potentiometer with electrodes at each end. The coil is quadruple, consisting of *A*, *B*, *C* and *D* of Figure 8/2/3.

Cybernetics

Sentience in terms of their outward manifestations not internal mechanism or subjective experience



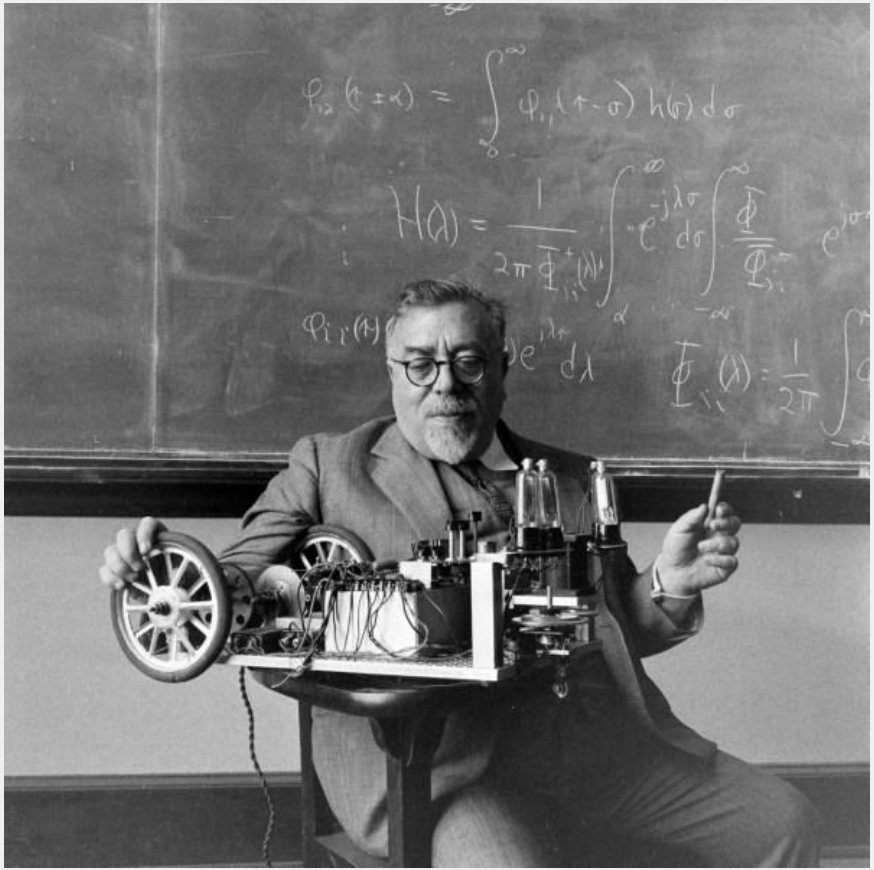


Elmer and Elsie 1948 (Grey Walter)



Voluntary Feedback

Light-seeking /
Light-avoiding
Voluntary Feedback

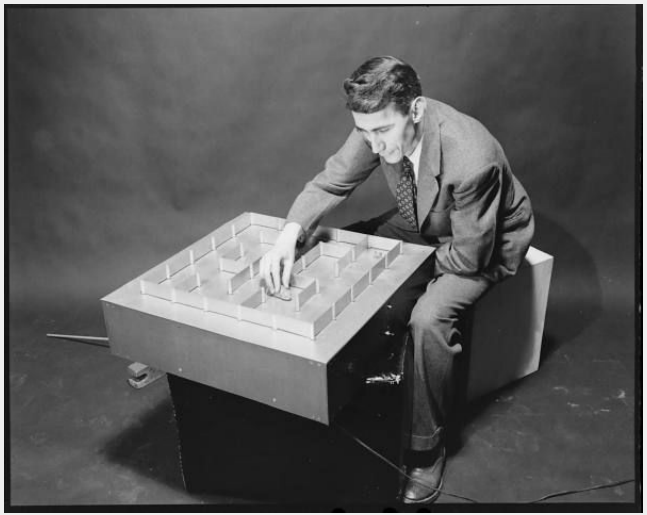


Moth/BedBug Machine

Early Reinforcement Learning

“Exploration Strategy” vs. “Goal Strategy”



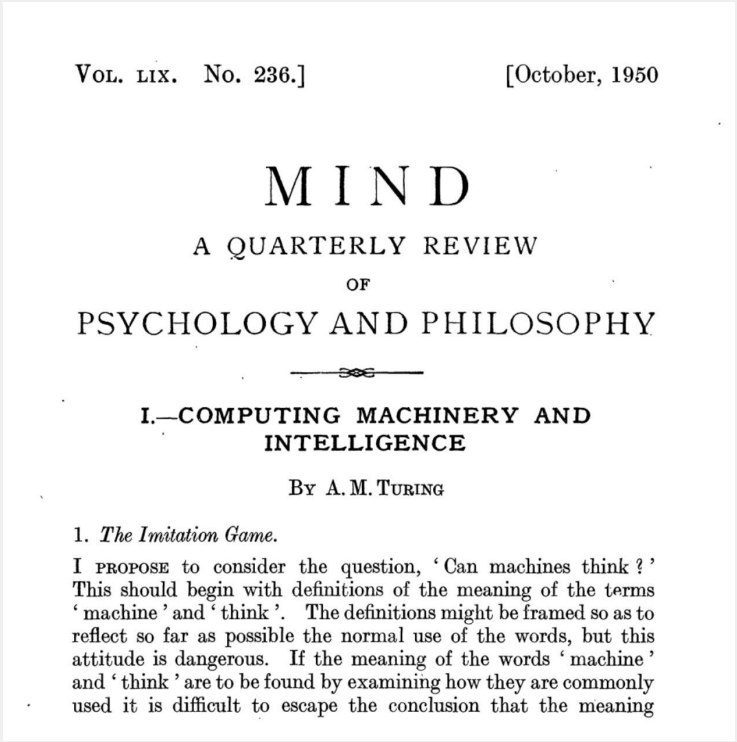


Claude Shannon’s Mouse Theseus 1950

Turing

Intelligence develops through learning

- 1950 “Computers Machinery and Intelligence”:
- Pleasure/Reward vs. Pain/Punishment
- “Machines must be fallible”
- Mathematical equations can’t determine whether a machine can think



VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

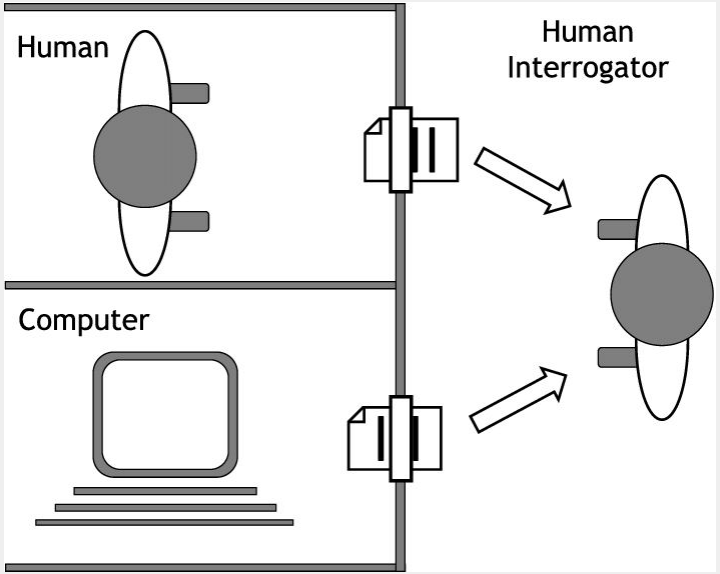
I PROPOSE to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

Turing

Appearing to have free-will

The triumphant machine's designers themselves would remain at least partially in the dark regarding its inner workings.147 This was necessarily the case, since to be able to describe the machine's inner workings completely would be to lose the sense that it was intelligent. But then, the same would be true in the case of a man: if one could attribute his behavior entirely to a set of explicit rules, Turing argued, one would come to see him as a machine and, by implication, without intelligence.

Riskin, Jessica. The Restless Clock: A History of the Centuries-Long Argument over What Makes Living Things Tick (p. 335).



Beginning of Modern AI

A Critical Mass of Researchers Fo
the “First Generation”



French Academie de Sciences 17th C



Dartmouth Conference 1956

“Modernizing” AI

“We propose that a 2 month, 10 man (**John McCarthy hosted; Minsky, Claude Shannon, Arthur Samuel, Ray Solomonoff, etc.**) study of **artificial intelligence** be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hamsphire.”

→ Stanford, CMU, MIT, IBM, etc.

Philosophy — reasoning, logic, rationalism
Mathematics – mathematical modeling, algorithms, probability
Economics – game theory, decision theory
Neuroscience – neurons, inspiration for computers
Psychology – cognitive science (brain as a information processing device)
Computer engineering – hardware and software
Cybernetics/Control Theory – homeostasis, objective function
Linguistics – knowledge representation, computational linguistics

- What are the characteristic differences between “early modern” and “modern” eras of Science and AI?
- What does “modernizing” a new field mean?
- Do you think this transition was inevitable?

- Looking purposeful vs. Being purposeful
- Degrees of Autonomousness:
  - Able to repeat tasks
  - Able to take varying input
  - Able to select input
  - Able to learn from experience

- What is Jessica Riskin’s main argument or POV?
- 2025 version of this book
- How would Riskin define AI?
- What is New Science?

Why AI?

THE U.S. Sun | News | Sport | Lifestyle | Entertainment | Money | Tech

A FORMER Google engineer who was let go by the tech giant and claimed its AI was sentient has made some more concerning claims.

Blake Lemoine made headlines last year after claiming Google's Large Language Model (LLM) had thoughts and feelings like a child.



Blake Lemoine is an AI consultant and public speaker who thinks some artificial intelligence could be sentient Credit: Getty

Google strongly denied these claims and said the AI was designed to respond like a human, but it's not sentient like us.

Lemoine has now published a new essay in Newsweek in which he discusses Microsoft's Bing's Open AI-powered chatbot.

The New York Times Account

A.I. Is Not Sentient. Why Do People Say It Is?

Robots can't think or feel, despite what the researchers who build them want to believe.

